



Pan-genome Analysis and Multilocus sequence typing

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Learning Objectives

By the end of this module, participants will be able to:

- ✓ Understand the theoretical concepts of pan-genomics and MLST.
- ✓ Gain practical experience with tools for pan-genome analysis (e.g., Roary) and MLST (e.g., mlst, PubMLST).
- ✓ Learn to interpret outputs and apply them to microbial genomics research.



Bacterial Comparative Genomics

- **Definition:** The analysis and comparison of complete genomes from different bacterial species, strains, or isolates.
- **Approach:** Aligns whole genomes or subsets (core vs. accessory genes) to identify similarities and differences.
- **Tools:** Involves genome sequencing, annotation, alignment, and visualization pipelines (e.g., Roary, Panaroo, Mauve, BRIG).
- **Key Focus:** Distinguishing core genome (shared by all strains) vs. accessory genome (strain-specific genes, plasmids, mobile elements).

Why Comparative Genomics of Bacteria?

- **Understand Evolution:** Reveals gene gain/loss, horizontal gene transfer, and evolutionary relationships.
- **Track Pathogens:** Identifies clonal lineages, outbreak sources, and transmission dynamics.
- **AMR & Virulence:** Detects antimicrobial resistance (AMR) genes and virulence factors across strains.
- **Adaptation & Niche Specialization:** Explains how bacteria adapt to hosts, environments, or stress conditions.
- **Vaccine & Drug Targets:** Highlights conserved genes or pathways suitable for therapeutics.



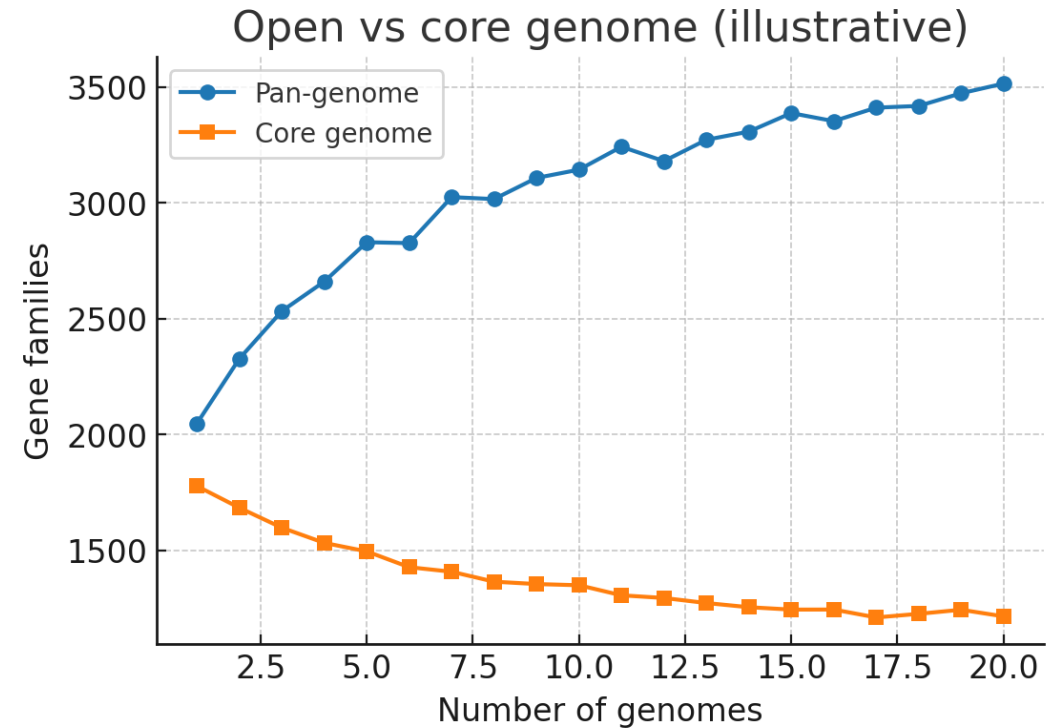
Pan-Genome vs Core Genome

Pan-Genome

- **Definition:** The entire gene repertoire found across all strains of a species.

Components:

- ▶ **Core genome** (shared by all strains)
- ▶ **Accessory genome** (present in some strains; niche adaptation)
- ▶ **Unique genes** (found in only one/few isolates)
- **Dynamic:** Expands as more genomes are sequenced.
- **Example:**
- AMR genes (e.g., blaCTX-M, blaKPC)



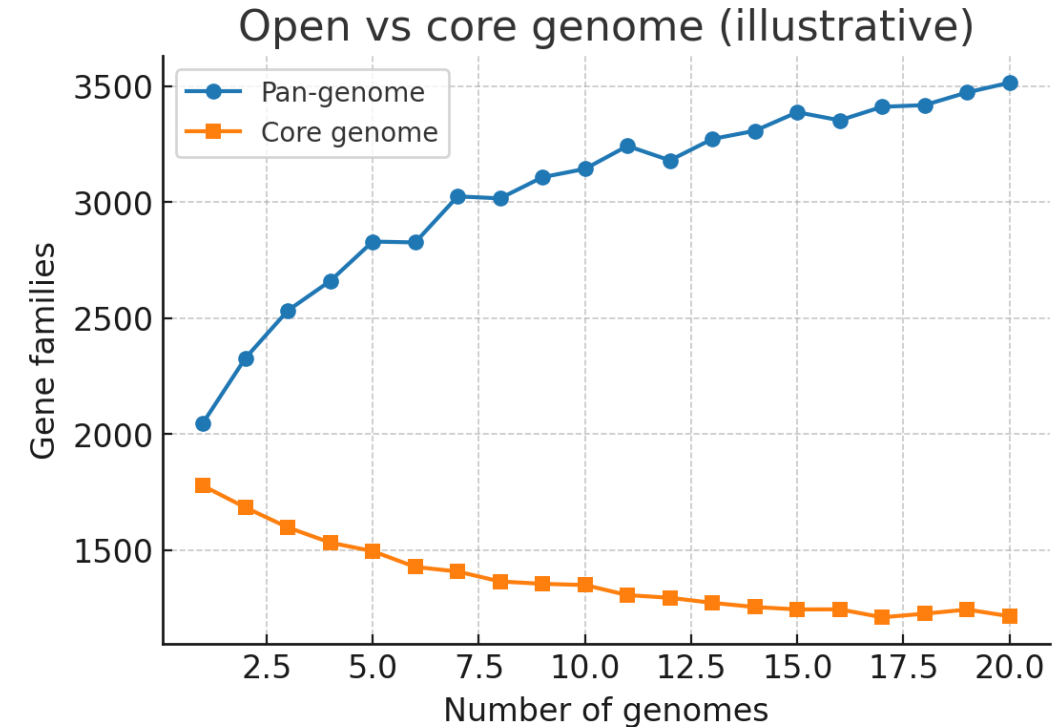
Pan-Genome vs Core Genome

Core Genome

- **Definition:** The set of genes present in all strains of a species.
- **Represents:** Essential functions required for survival (e.g., DNA replication, transcription, translation, basic metabolism).
- **Stable:** Highly conserved; changes slowly over evolutionary time.

Example:

- In *E. coli* and *Klebsiella*, the core includes housekeeping genes like *gyrA*, *rpoB*, *recA*.
- Forms the backbone for species identity.

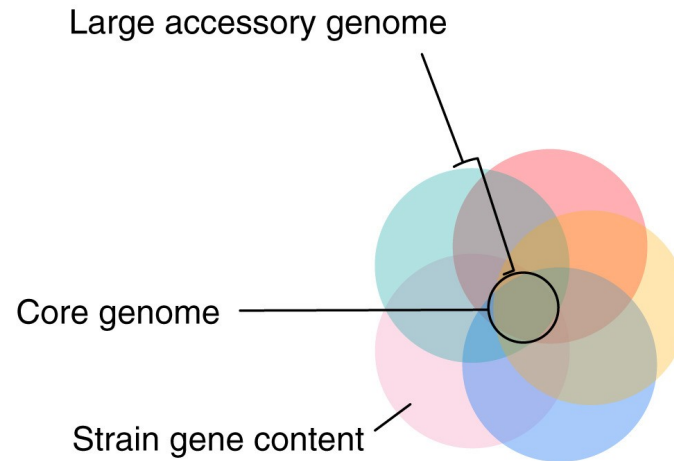


Pan-genome Evolution Models

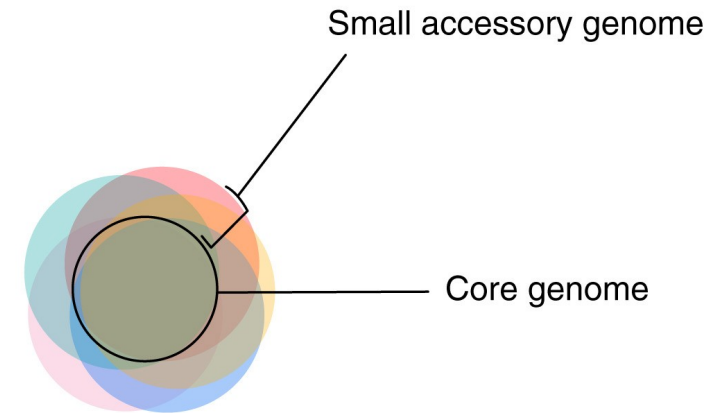
- **Open Pan-genome:**

- Continuous discovery of new genes as more genomes are added
- Common in highly diverse, environmental bacteria

Open pangenomes



Closed pangenomes



- **Closed Pan-genome:**

- Limited new genes with addition of more genomes
- Typical of species with restricted habitats

Common among....

Niche generalists
Diverse community interactions
Large population size



Niche specialists
Limited community interactions
Small population size

Current Biology

Tools for Pan-genome Analysis

Tools

- **Roary:** rapid pan-genome analysis from annotated genomes
- **Panaroo:** robust handling of errors in assemblies/annotations
- **BPGA:** Bacterial Pan Genome Analysis pipeline
- **Anvi'o:** visualization-rich pan-genomics platform

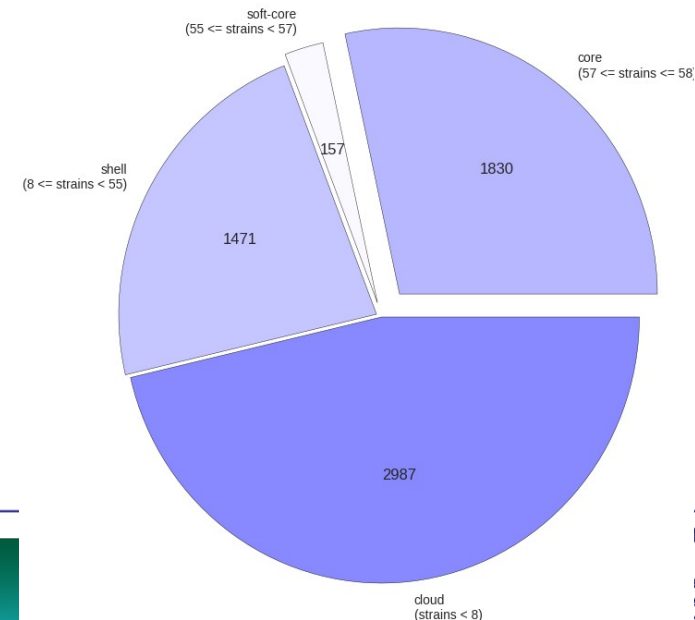
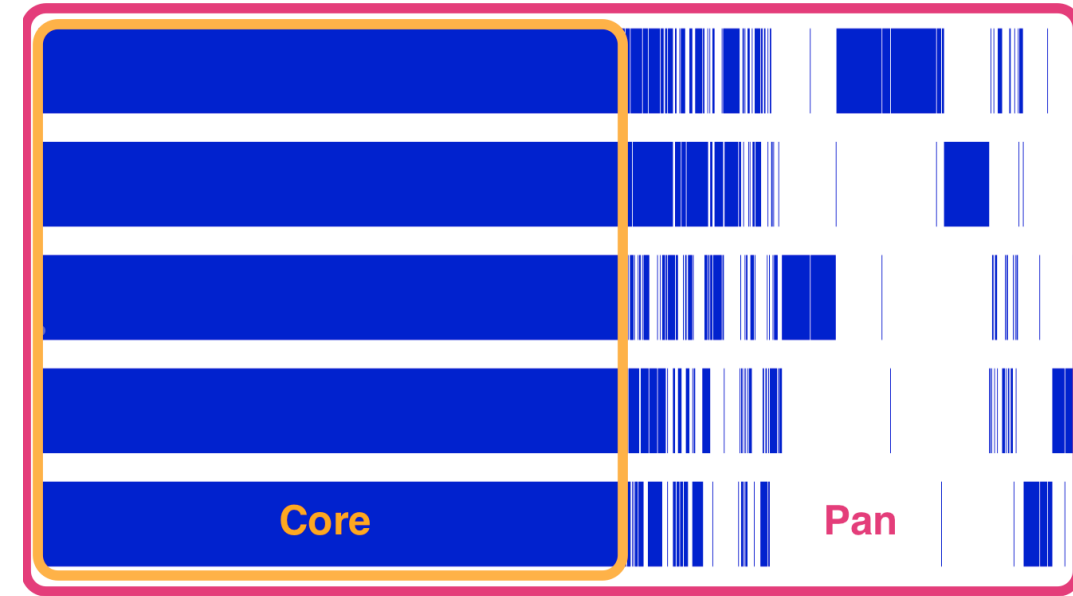
Roary Workflow

1. Annotate genomes using Prokka
2. Run Roary with GFF3 files
3. Generate gene presence/absence matrix
4. Build core gene alignment
5. Visualize phylogeny and accessory genome variation

Pan-genome Analysis

Roary Output

- **gene_presence_absence.csv:** A CSV file showing the presence or absence of each gene cluster in each sample.
- **Core Gene Alignment:** Alignments of the core genes, which are crucial for downstream analyses like SNP identification and phylogenetic tree construction.
- **summary_statistics.txt:** A text file summarizing the number of core and accessory genes found across the dataset.



Multi-locus sequence typing (MLST)

- It relies on sequence variation in a defined set of housekeeping genes to classify related strains (sequence types, STs) and related groups of STs (clonal complexes).
- **Housekeeping genes used in the E. coli MLST scheme:** The standard Achtman E. coli MLST scheme targets ~450–500 bp internal fragments of seven housekeeping genes: **adk** (adenylate kinase), **fumC** (fumarate hydratase), **gyrB** (DNA gyrase subunit B), **icd** (isocitrate dehydrogenase), **mdh** (malate dehydrogenase), **purA** (adenylosuccinate synthetase), and **recA** (recombinase A).
- **Diversity captured in E. coli:** By 2020, more than 15,000 MLST sequence types (STs) had been recognized for E. coli, reflecting the species' high genetic diversity and widespread use of MLST in clinical and environmental isolates.
- **Scheme definition:** Like in other bacteria, the E. coli MLST scheme defines allelic variation by comparing nucleotide segments from each of the seven housekeeping genes, with each unique allelic profile assigned a distinct ST.

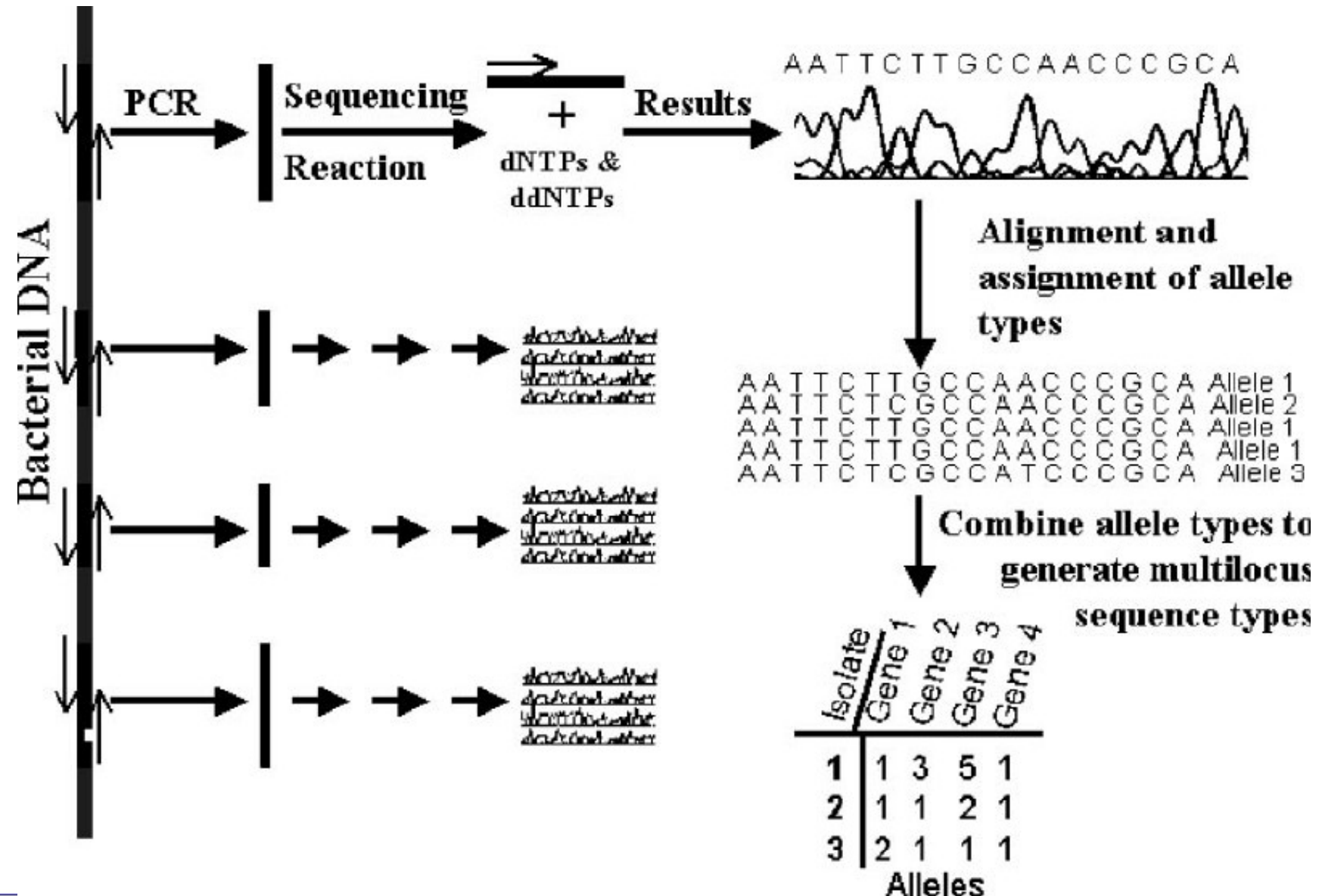
Multi-locus sequence typing (MLST)

- **MLST** is considered the “gold standard” of typing
- It was first developed for *Neisseria meningitidis* in 1998 to ensure reproducibility
- The principle behind the MLST scheme is to identify internal nucleotide sequences of approximately 400 to 500 bp in multiple housekeeping genes
- Unique sequences (alleles) are assigned a random integer number and a unique combination of alleles at each locus, an “allelic profile,” specifies the sequence type (ST)
- The MLST allele sequences and ST profile tables are stored in curated databases hosted at different sites around the world ([<http://pubmlst.org>])
- As of 2011, 66 bacterial MLST schemes were available, most of them function at the species level, e.g., *Escherichia coli* and a few function on the genus level, e.g., *Neisseria* schemes

Multi-locus sequence typing (MLST)

Methodology

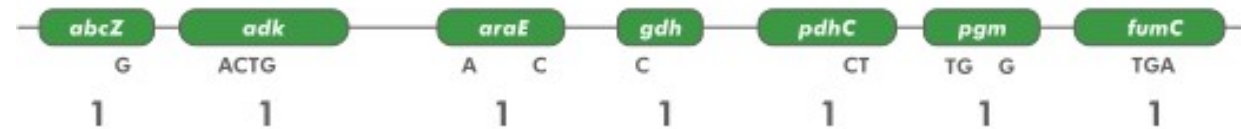
- Traditionally, MLST starts with a PCR amplification step using primers that are specific for the loci of the MLST scheme
- Followed Sanger sequencing



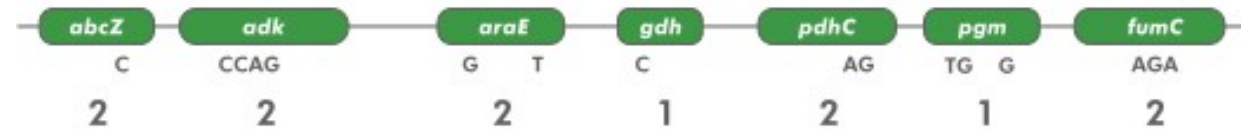
Multi-locus sequence typing (MLST)



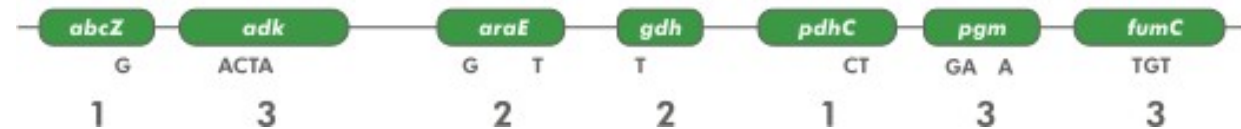
Strain A



Strain B



Strain C



- **Allelic assignment:** Each unique nucleotide sequence at a locus is given an allele number (e.g., adk = allele 1, 2, or 3). Identical sequences share the same allele designation.
- **Strain differentiation:**
 - Strain A has allelic profile: 1 1 1 1 1 1 1
 - Strain B has allelic profile: 2 2 2 1 2 1 2
 - Strain C has allelic profile: 1 3 2 2 1 3 3
- **Sequence type (ST) definition:** The allelic profile across all seven genes is combined into a ST. Strains A, B, and C therefore represent three different STs.
- **Population genetics insight:** By comparing allelic profiles, MLST allows researchers to define related strains, track clonal complexes, and study evolutionary or epidemiological relationships in *E. coli*.

Question Marks in MLST Analysis – What They Mean

- **"?" = Unknown / Missing Allele:** Sequence does not match any allele in the MLST database.
- **Novel or Ambiguous Data:** Could represent a new allele or poor-quality sequencing.
- **Unassigned Sequence Type (ST):** If one or more loci = "?", the isolate cannot be assigned to a known ST.
- **Missing Gene Segment:** Locus may be absent or not recovered due to sequencing/assembly gaps.
-  Researchers can submit **novel alleles** → assigned official allele number & new ST

1	2	3	4	5	6	7	8	9	10
dataset_345341.dat	saureus	36	arcC(2)	aroE(2)	glpF(2)	gmk(2)	pta(3)	tpi(3)	yqiL(2)

scheme

sequence type

genes and their alleles

Multi-locus sequence typing (MLST) vs WGS

Multi-locus sequence typing	Whole genome sequencing
Compares organisms based on genes in the MLST scheme for that organism	Enables comparison of organisms at a whole genome level
Marker genes pre determined	Challenging to readily discern comparative markers from large data without a globally standardized informatic pipeline
Traditionally performed in an expensive and time-consuming manner	Faster and cheaper with continuous decline in cost of WGS
Easier to interpret results	More demanding to extract the relevant information from the large amount of data
Resolution for some organisms stops at genus level	Resolution upto subspecies level

Multi-locus sequence typing (MLST) vs WGS

Sequencing platform	Species	No. of isolates	MLST scheme
Illumina			
Paired-end reads	<i>B. animalis</i>	5	<i>Bifidobacterium</i>
	<i>C. jejuni</i>	53	<i>Campylobacter</i>
	<i>E. coli</i>	15	<i>E. coli</i> scheme 1
	<i>E. coli</i>	15	<i>E. coli</i> scheme 2
	<i>K. pneumoniae</i>	4	<i>K. pneumoniae</i>
	<i>L. lactis</i>	34	<i>L. lactis</i>
	<i>S. aureus</i>	83	<i>S. aureus</i>
	<i>S. enterica</i>	50	<i>S. enterica</i>
	<i>S. thermophilus</i>	13	<i>S. thermophilus</i>
	Single reads	<i>E. coli</i>	<i>E. coli</i> scheme 1
		<i>B. animalis</i>	<i>Bifidobacterium</i>
		<i>B. longum</i>	<i>Bifidobacterium</i>
		<i>L. lactis</i>	<i>L. lactis</i>
		<i>P. aeruginosa</i>	<i>P. aeruginosa</i>
		<i>S. thermophilus</i>	<i>S. thermophilus</i>
Roche 454	<i>E. coli</i>	3	<i>E. coli</i> scheme 1
	<i>P. aeruginosa</i>	2	<i>P. aeruginosa</i>
Ion Torrent	<i>E. coli</i>	2	<i>E. coli</i> scheme 1
SOLiD	<i>E. coli</i>	1	<i>E. coli</i> scheme 1

Core genome Multilocus Sequence Typing(cgMLST)

- cgMLST extends traditional MLST, which typically analyzes 7 loci, by examining a larger set of conserved genes (the "core genome") across different bacterial isolates.
- This approach provides higher resolution for differentiating strains, making it useful for;
 - Microbial epidemiology
 - Phylogenetics
 - Tracking outbreaks
- **Procedure**
 1. A set of core genes that are conserved across all members of a bacterial species.
 2. 100 or even 1000 of genes are chosen to represent the core genome and are identified from WGS using comparative genomics tools - cgMLST target definer.

Core genome Multilocus Sequence Typing(cgMLST)

- **cgMLST target definer (version 1.4)**
- Input annotated WGS from databases such as NCBI
- BLAST is used to identify gene clusters
- Genes presented in nearly all members of the species are then selected (core genes)
- The selected loci should be stable, meaning they experience low rates of recombination and mutation
- Export the defined core loci and their allele numbering to create a functional cgMLST scheme
- **Enterobase** - has schemes for cgMLST for *E.coli*, *Salmonella*, *Shigella*, *MTB*, *Vibrios*

Pan-genome vs MLST

Differences

MLST:

- Based on a small set of housekeeping genes
- Useful for typing and epidemiological tracking
- Easy to compare across labs via PubMLST

Pan-genome:

- Uses the entire gene repertoire of genomes
- Provides insights into evolution, adaptation, and gene flow
- Rich data but requires more computational resources

When to use what!!

- **MLST** provides rapid, standardized strain typing
- **Pan-genomics** offers detailed insights into genetic diversity

Together:

- Trace outbreaks (**MLST**)
- Understand mechanisms of resistance and adaptation (**Pan-genome**)
- Guide public health interventions and vaccine design

Key Takeaways

- Pan-genome = core + accessory + unique genes
- MLST = fast, standardized typing method
- Both methods complement each other in microbial genomics
- **Applications:** AMR, surveillance, outbreak investigations, evolution
- Essential for integrating genomics into public health and research